# Derick Amalraj

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## SUMMARY

Senior Software Development Engineer focused on build, test, and CI infrastructure for hardware and firmware validation. Own the CI pipelines, simulation runner fleets, and test scheduling that large engineering teams depend on, with an emphasis on fast, trustworthy feedback, failure triage and bisection, observability, and developer productivity. Experienced in influencing cross-functional hardware, firmware, and software teams to adopt shared quality standards and self-service tooling at scale.

## SKILLS

- **Build, Test & CI/CD:** GitHub Actions, Jenkins, Groovy, pre-merge/merge/nightly pipelines, test scheduling, failure triage, regression bisection, release-readiness reporting, developer productivity
- **Fleet & Infrastructure:** Runner fleet management, resource scheduling, Kubernetes, Docker, Proxmox, virtualization, custom OS images, Terraform, Helm, Linux
- **Observability:** Grafana, Elasticsearch, CI health dashboards, KPI and validation analytics
- **Programming:** Python, C/C++, Unix shell, CUDA/ROCm, Java, JavaScript, TypeScript, SQL, GraphQL, MongoDB
- **Leadership:** Technical leadership, cross-functional collaboration, stakeholder communication, quality governance, process improvement

## RELEVANT EXPERIENCE

**Senior Software Development Engineer, Advanced Micro Devices (AMD)** **July 2026 - Present**

- Own build, test, and CI for firmware validation across multiple firmware components and hardware programs, providing readiness, reliability, and regression visibility to engineering leadership.
- Manage runner fleets and resource scheduling for simulation-based hardware testing, including virtual machine capacity, virtualized test environments, and custom operating system images, keeping shared compute reliable and well utilized.
- Reduced simulation execution time by over 70% (2 hours $\rightarrow$ 15 minutes) through resource scheduling improvements, test restructuring, and virtual machine capacity optimization.
- Expanded pre-merge and nightly validation coverage across BIOS, platform, power-management, and security firmware, enabling earlier defect detection while maintaining validation stability and throughput.
- Triage validation failures and bisect regressions to identify the change or point of failure, shortening time to root cause and restoring a trustworthy signal for engineers.
- Built Grafana dashboards, KPI reporting, and validation analytics to track test stability, failure trends, validation coverage, and CI reliability across server, client, and accelerator firmware programs.
- Developed automated quality reports summarizing 30–45+ validation jobs per execution, improving regression visibility, infrastructure diagnostics, and release-readiness assessment for engineers and leadership.
- Improve developer productivity by refactoring the internal CI infrastructure for modularity and reusability, consolidating previously incompatible code to accommodate growing testing needs and inbound requests.
- Coordinated organization-wide rollout of validation requirements and quality gates across firmware repositories, and designed a centralized communication model supporting over 450 engineers and managers.
- Designed scalable CI support and onboarding processes that improved self-service, supportability, and cross-team coordination.
- Automated simulation-based validation workflows with Python and shell scripting to verify firmware quality across multiple hardware platforms.
- Received 3 AMD Spotlight awards for infrastructure management and post-silicon/onboard testing.

## ADDITIONAL EXPERIENCE

- **Software Development Engineer 2, Advanced Micro Devices (AMD)** **April 2024 - June 2026**
- **Teaching Assistant, University of Toronto** **January 2023 - December 2023**
- **Software Engineer Intern, Advanced Micro Devices (AMD)** **May 2021 - August 2022**

## EDUCATION

- **Master of Science in Applied & Computational Mathematics, University of Washington** **Spring 2026**
- **Bachelor of Science in Computer Science & Mathematics with Honours, University of Toronto** **Summer 2023**